

GEOGRAPHY

PHYSICAL GEOGRAPHY

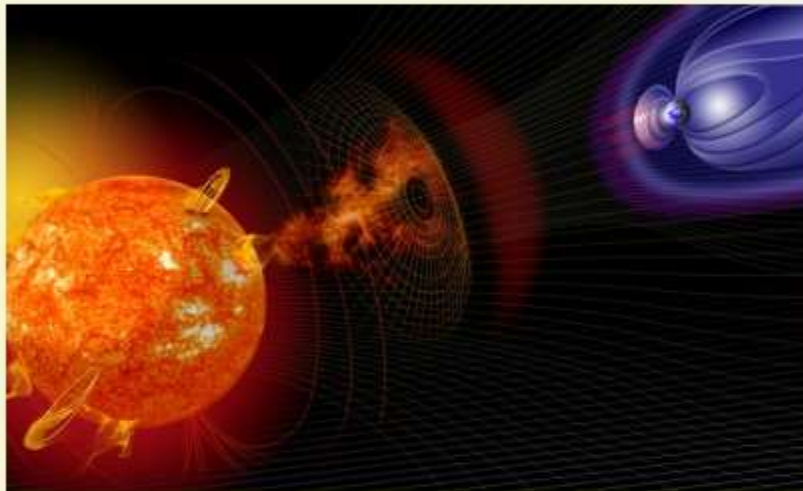


MODULE -6

- HOW SOLAR FLARES AFFECT EARTH?
- WHAT IS SOLAR WIND?
- WHAT CAUSES AURORA BOREALIS?
- WHY MANY COLOURS IN AURORAS?



How the solar flares affect the Earth?

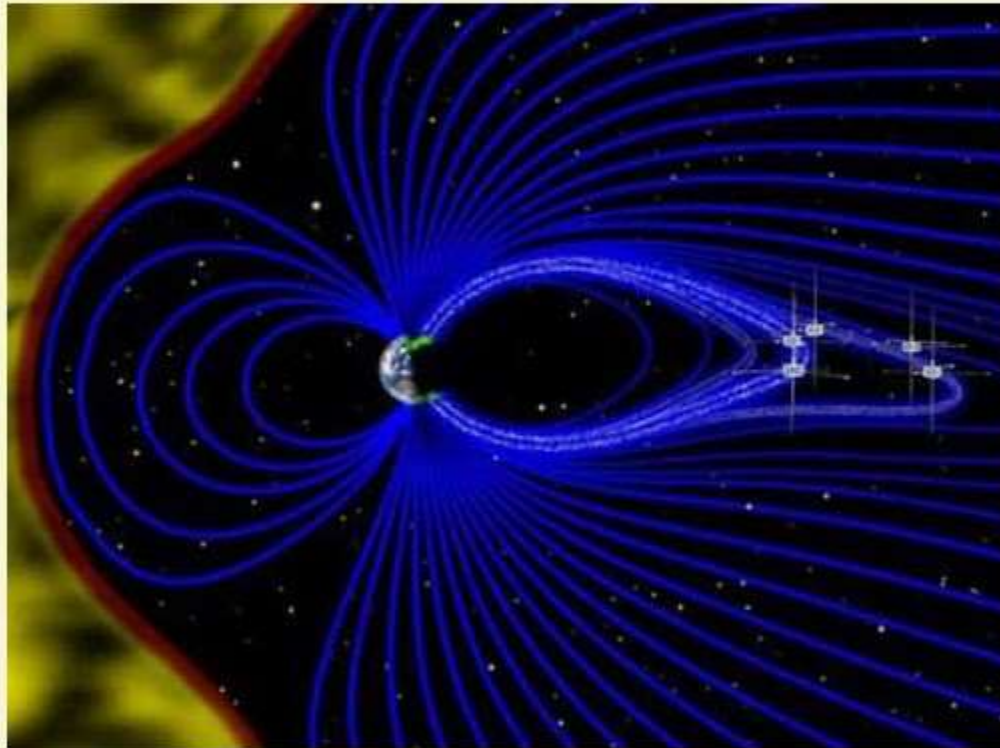


Everyone of us is shielded by the Earth's atmosphere and magnetic field. The radiation is over the whole light spectrum. But, we are mostly concerned with x-rays and gamma rays, since they are the most energetic.

- It can disrupt communication networks and navigation systems on Earth.
- Astronauts in space risk being hit by these hazardous particles.
- Manned missions to Moon or Mars must take this danger into account.
- Sensitive electronic equipment in space can also be damaged by this energetic particles.
- Sometimes solar flares are associated with Coronal Mass Ejections (CMEs).
- CMEs result in the ejection of highly energetic protons, electrons and ions.



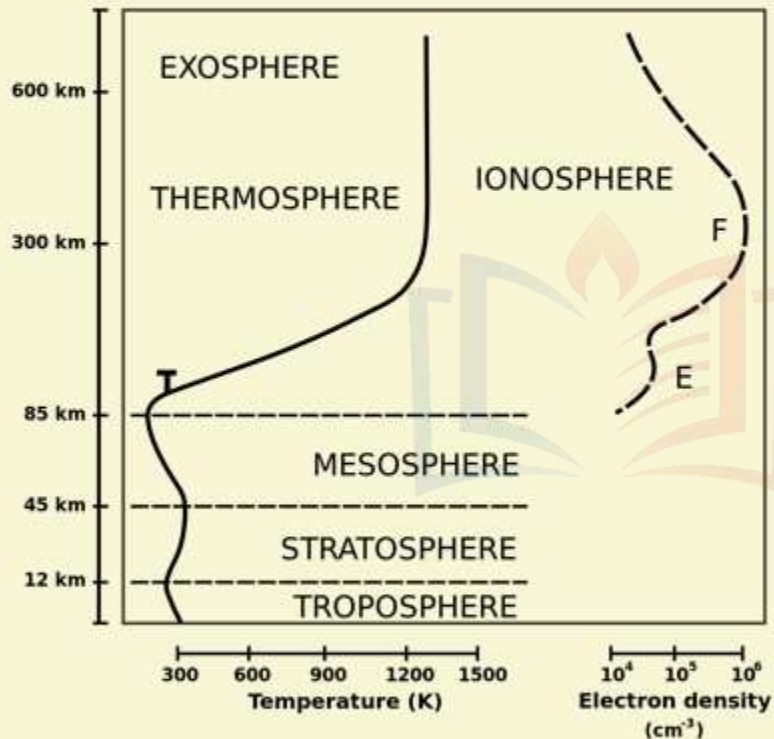
How the solar flares affect the Earth? (... contd)



- All of these can cause some trouble for airline pilots.
- Normally, there is natural protection in the form of Earth's magnetic field and atmosphere. But, the magnetic field at poles is weak.
- Hence, if the pilot flies nearer to the pole, he may get struck by these charged particles.
- The charged particles, if they hit the electric systems of plane can cause disruptions.



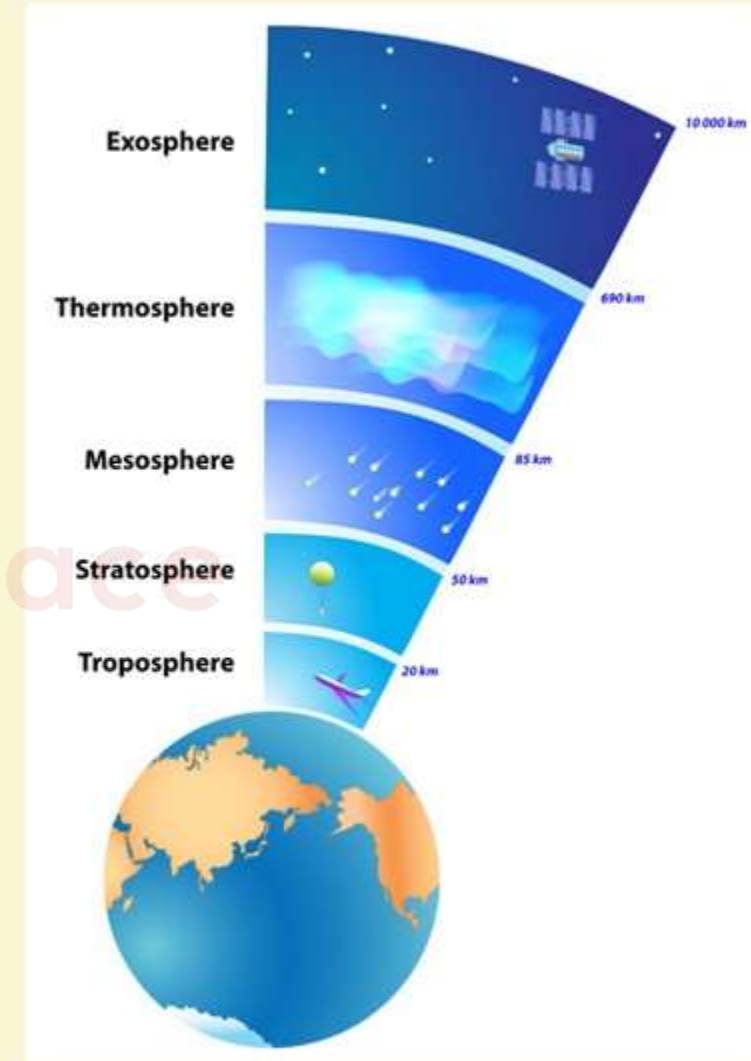
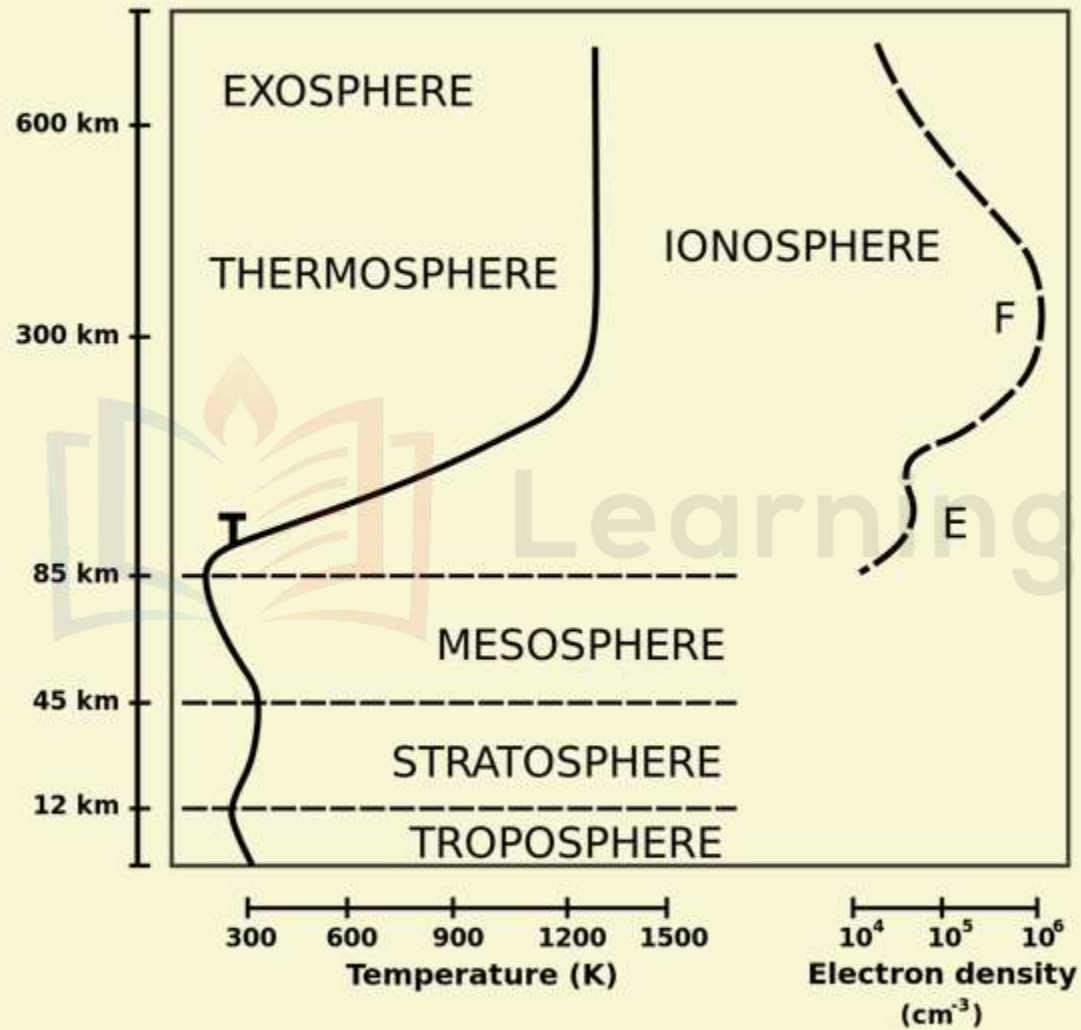
How the solar flares disturb radio communication and affect satellites?



- 1 The increase in heat and energy results in expansion of the Earth's Ionosphere.
- 2 Manmade radio waves travel through this portion of the Earth's upper atmosphere.
- 3 So, radio communication can be disturbed by its sudden unpredictable expansion.
- 4 Similarly, satellites circling through vacuum-free space can find themselves caught in the expanded sphere.

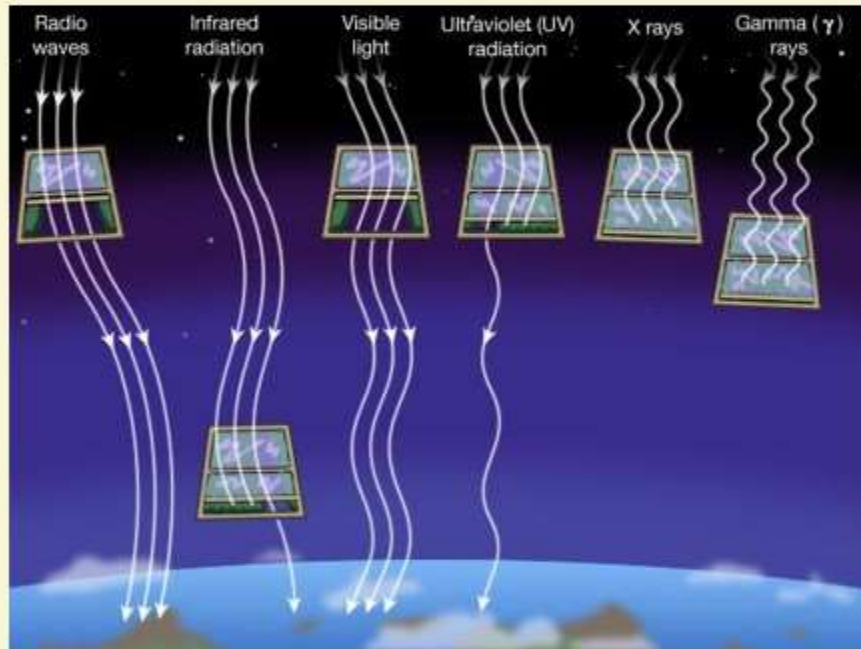
The ionosphere is the ionized part of Earth's upper atmosphere, from about 60 km to 1,000 km altitude, a region that includes the thermosphere and parts of the mesosphere and exosphere.







How the solar flares disturb radio communication and affect satellites? (... contd)



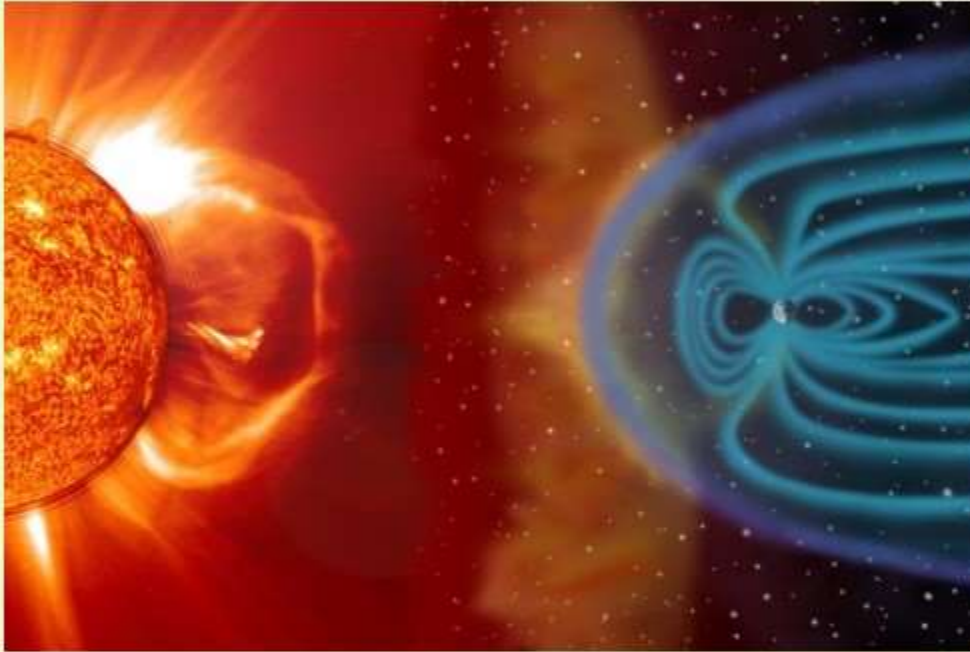
5 The resulting friction slows down their orbit

6 Since x-rays and gamma rays failed to penetrate the atmosphere, only space-based telescopes can detect their signatures.

Solar flares are almost never visible optically. The bright emission of the surrounding photosphere, where the Sun's light originates overshadow this explosive phenomena.



What do you understand by solar wind?



The Sun continuously emits electromagnetic radiation and highly energised particles into space which produce space weather. Solar wind is part of space weather.

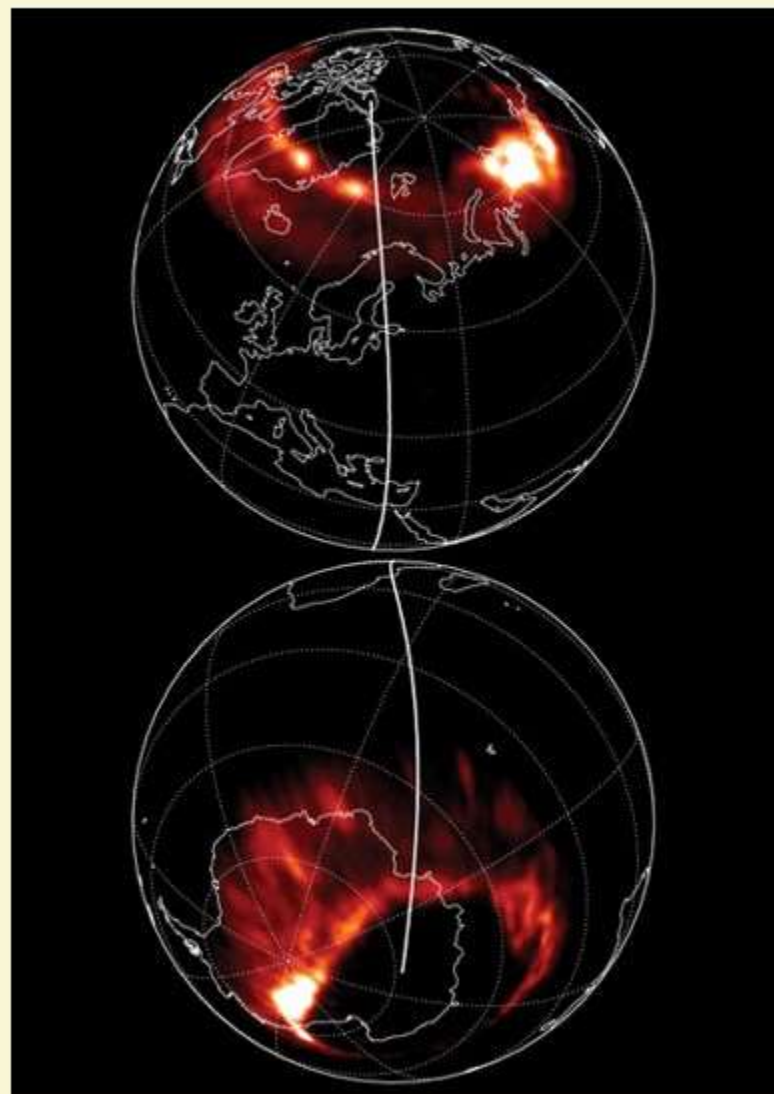
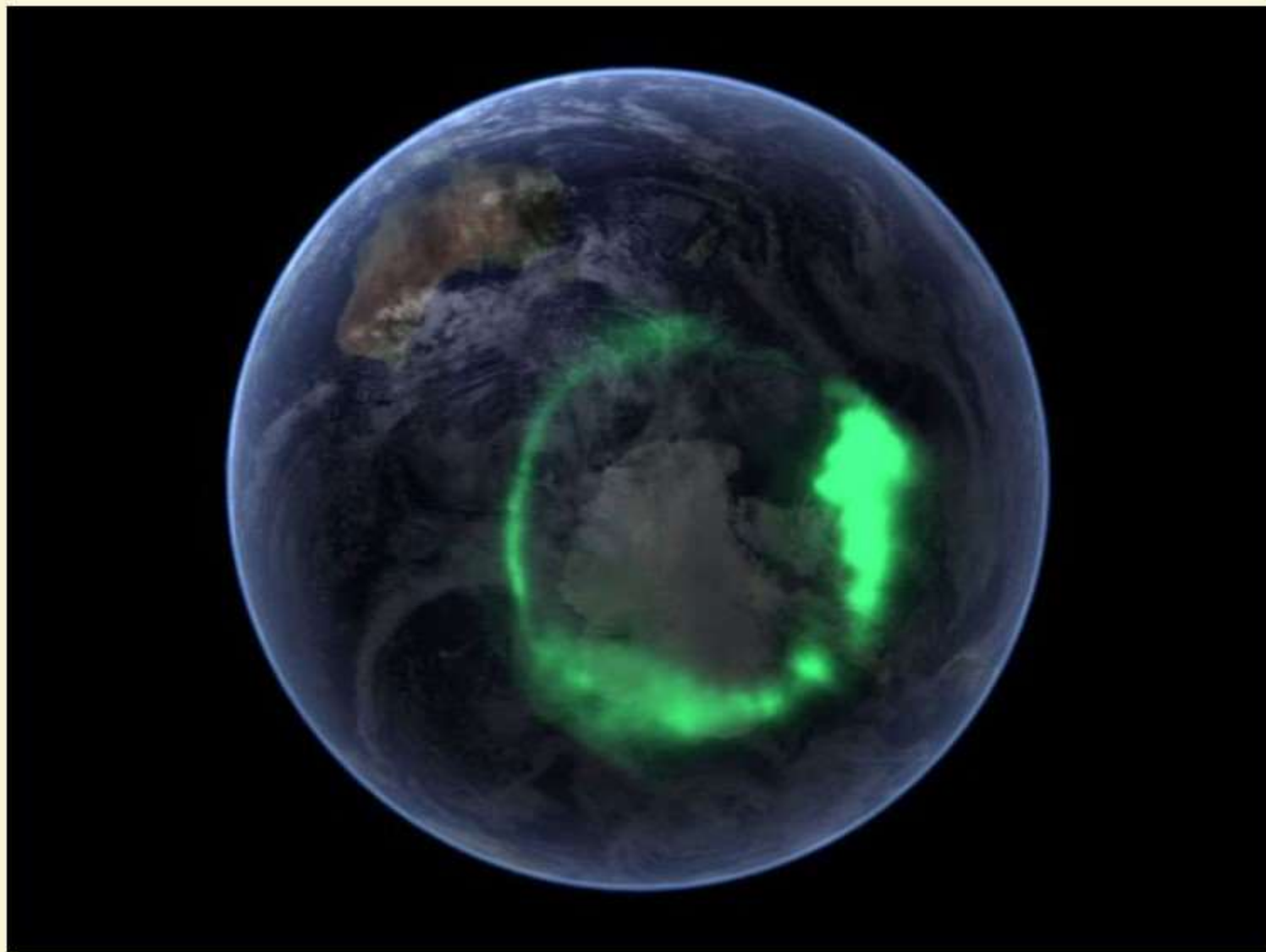
- The solar wind is a continuous stream of highly energised particles, mostly electrons and protons.
- They flow out from the Sun, through space at very high speeds and high temperature.
- They travel at speeds as high as 900 km/s and at a temperature of around 1 million degrees (Celsius).
- It is made of plasma.
- The solar wind is caused by the hot solar corona, which is the outermost layer of the solar atmosphere.



What causes Aurora Borealis?

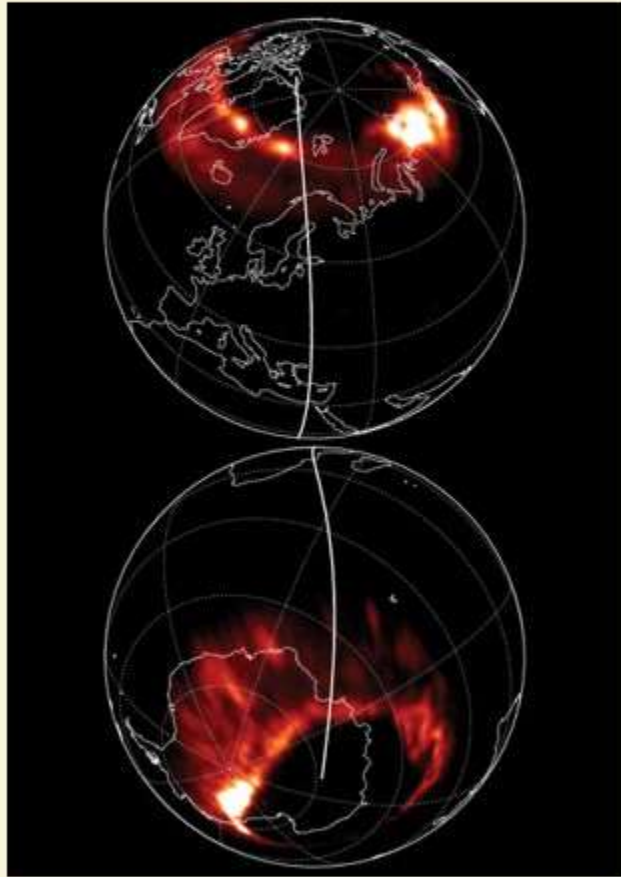


- We have just now seen that particles escape from the Sun, known as solar wind into space.
- The temperature above the surface of the Sun is millions of degree Celsius.
- At this temperature, collisions between gas molecules are frequent and explosive.
- Free electrons and protons are thrown from the Sun's atmosphere and blown towards the Earth by the solar wind.
- The charged particles are largely deflected by the Earth's magnetic field.





What causes Aurora Borealis?



- However, the Earth's magnetic field is weaker at either pole and some particles enter the Earth's atmosphere and collide with gas particles.
- These collisions emit light that we perceive as the dancing lights of the North and the South.
- These can be seen in the northern or southern hemisphere in an irregularly shaped oval centred over each magnetic pole.
- The lights are known as **Aurora Borealis** in the **North** and **Aurora Australis** in the **South**.



Aurora
Borealis



Aurora Australis





What causes Aurora Borealis?

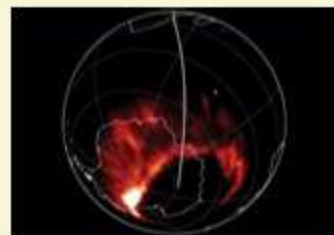


NORTHERN LIGHTS & SOUTHERN LIGHTS

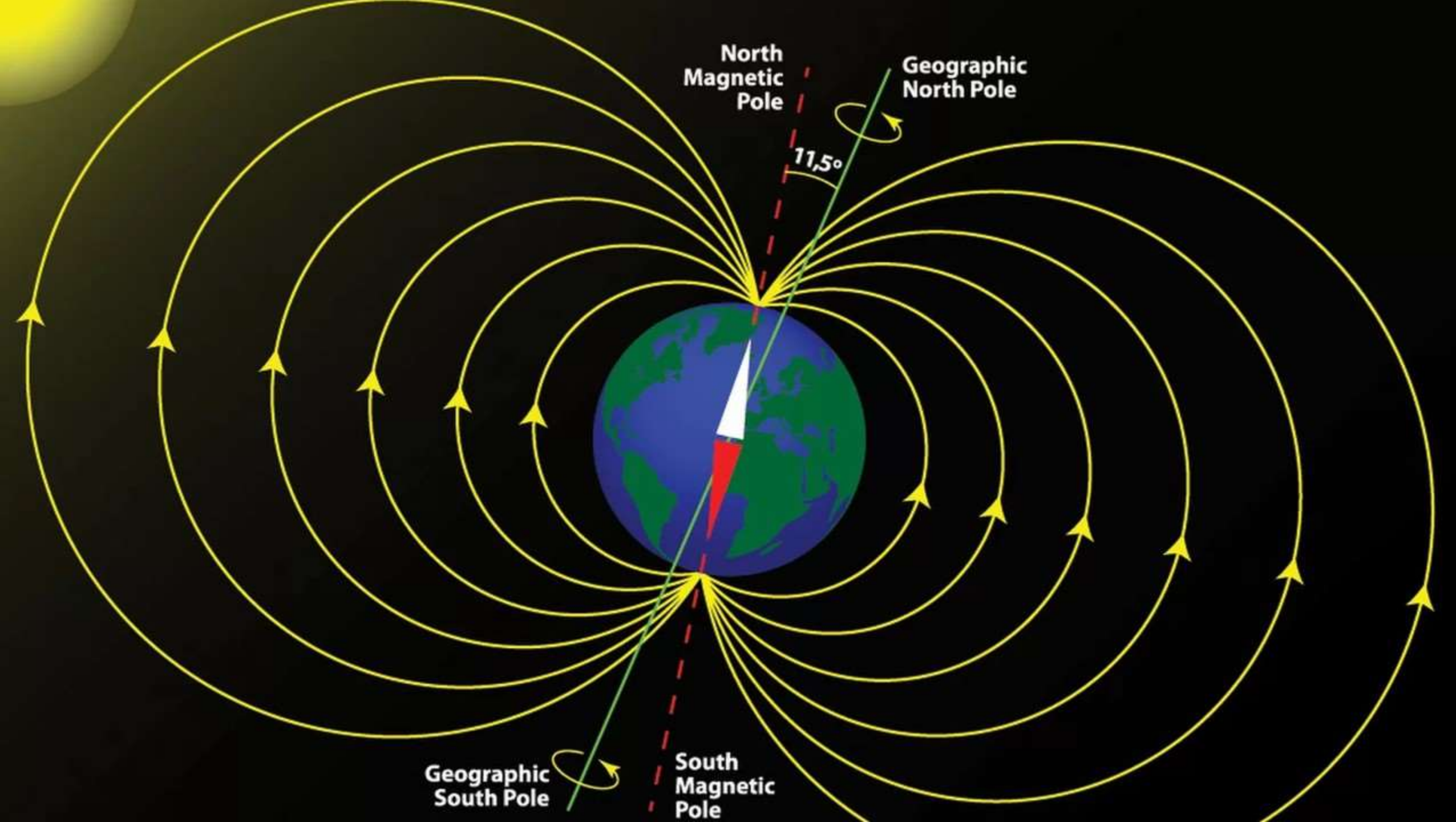
- 1** It is the natural electric phenomenon that creates bright and colourful light displays in the sky.
- 2** They are common at higher latitudes, mostly within Arctic and Antarctic circles.
- 3** In Arctic circle, it is known as Aurora Borealis and in the Antarctic circle, this is known as Aurora Australis.

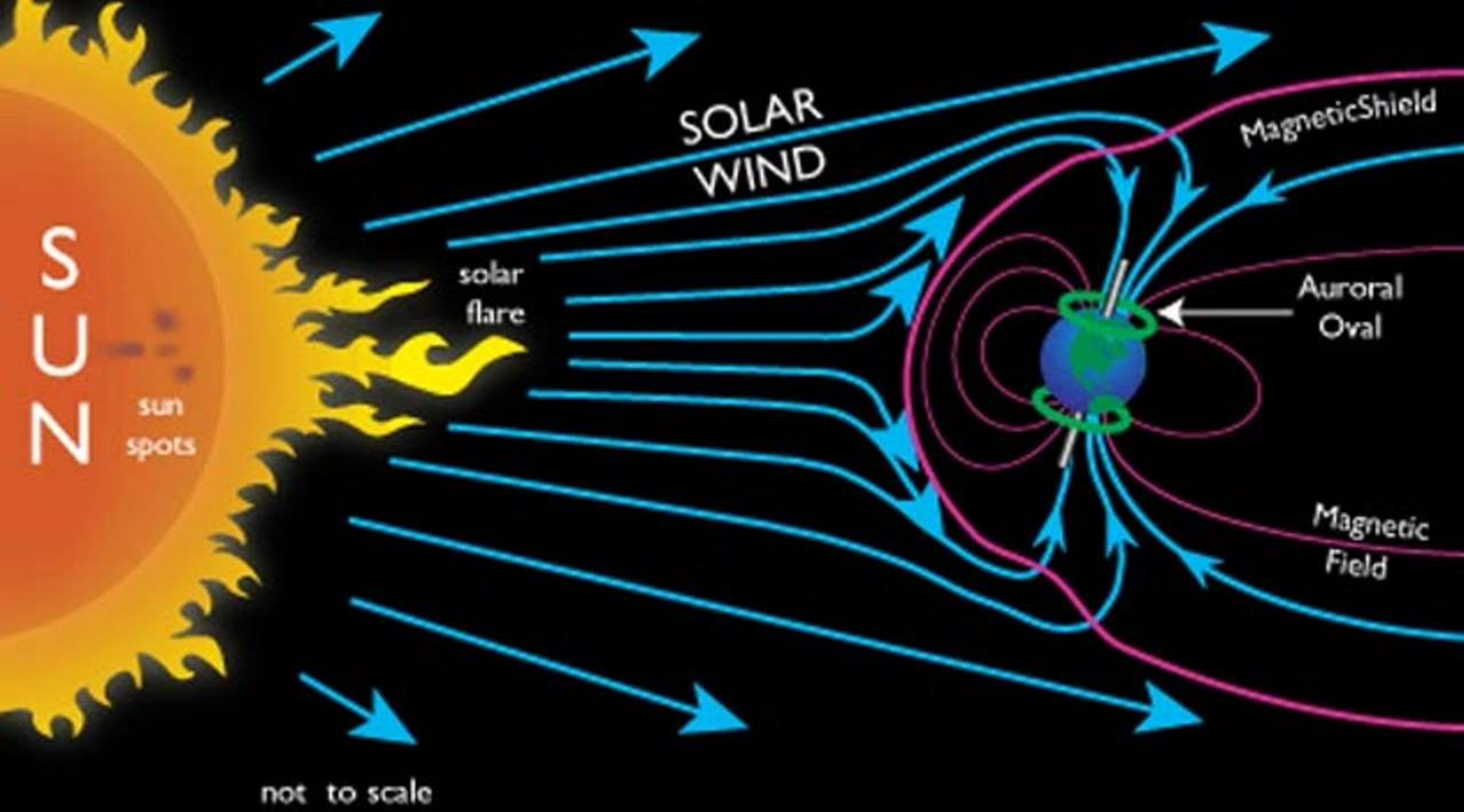


Why Aurora Borealis and Aurora Australis occur only at higher latitudes?



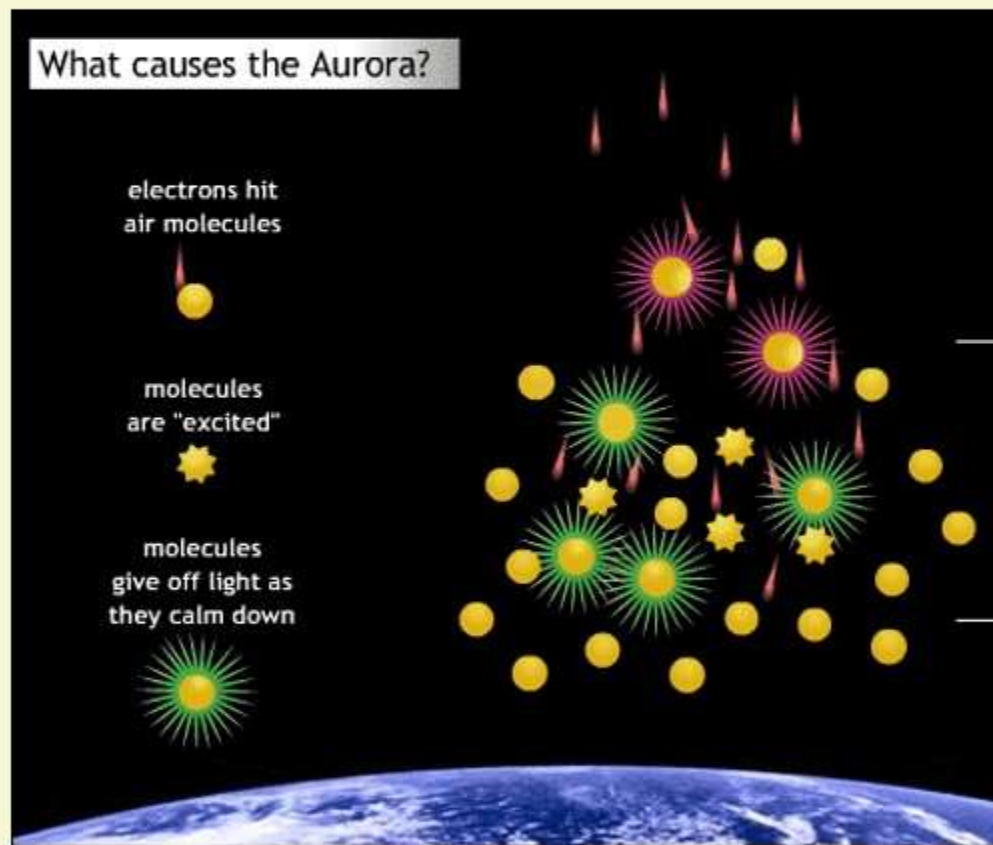
- 1** Earth is a giant planet, its magnetic field extends from the Earth's core to the area in space.
- 2** The region of the magnetic field, where the Earth's magnetic influence dominates is known as magnetosphere.
- 3** The Earth's magnetosphere shields the Earth from solar winds and other harmful cosmic rays.
- 4** It deflects most of the highly charged particles from solar winds and stops them from entering the Earth's atmosphere.
- 5** The Earth's magnetic field is weaker at either pole and some particles enter the Earth's atmosphere and collide with gas particles. This is what causes Northern lights and Southern lights.







What is the mechanism of this collisions?



- 1** The energy from the electrons in the solar winds is transferred to electrons in the atoms of different atmospheric gases.
- 2** Any excess energy is then released by these excited atoms in the form of light.
- 3** Aurora light displays tend to occur at between 50 miles to 200 miles above the surface of the Earth (these are tentative heights).



Why there are different colours?



- 1** The colour of the light released depends on the kind of gas molecules, their electrical state at the time of collision and the type of solar wind particles they collide with.
- 2** Oxygen atoms emit yellow, green or red coloured light, while nitrogen atoms generate blue or purplish-red coloured light.
- 3** A mix of gases in the Earth's atmosphere creates multi-coloured auroras.
- 4** As the solar winds continuously enter the Earth's atmosphere and interact with gas atoms, aurora displays can be static as well as dynamic. They can change shape, colours and pulsate in the skies.

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For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao

Amrita Naidu

D. Sunil Kumar

L. V. Krishna

D. Chaitanya

K. Srikanth

G. Ravi Babu

D. Joji

K. Victor Babu

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